Homework 8

Week 10

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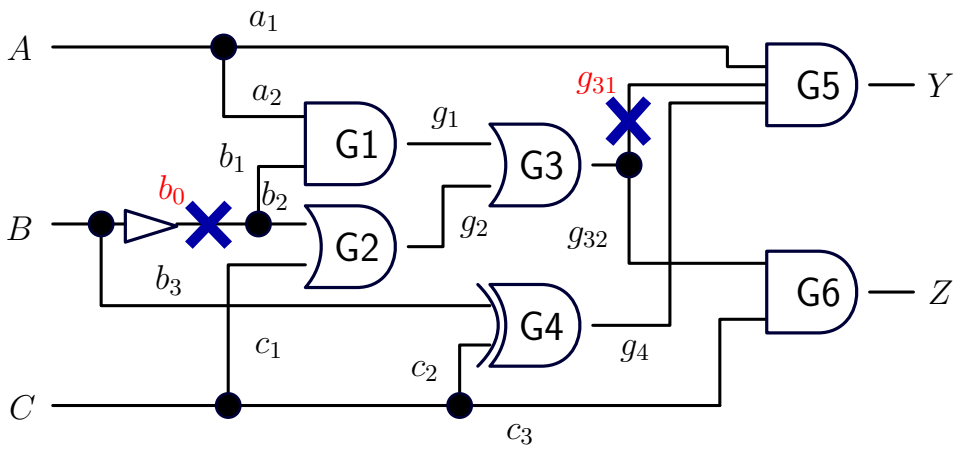

Problem 1. Find the test input when b_0 is stuck-at 0 fault.

Solution

Circuit functions:

$$\begin{aligned} G1 &= AB \\ G2 &= B + C \\ G3 &= AB + B + C = B + C \\ G4 &= B \oplus C \\ Y &= A \cdot G3 \cdot G4 \\ Z &= G3 \cdot C \end{aligned}$$

- **Activate:** b_0 s-a-0 $\Rightarrow$ need $B = 1$.
- **Propagate:** through G2 needs $c_1 = C = 0$ (then $G3$: good 1, faulty 0). Through G5 needs $A = 1$ and $g_4 = B \oplus C = 1$ (true for $B = 1, C = 0$). (Through G6 is impossible since $C = 0$.)

$$\boxed{ABC = 110} \quad \Rightarrow \quad Y_{good} = 1,\ Y_{faulty} = 0.$$

Problem 2. How many faults can be tested by the test input on 1?

Solution

Good values under $ABC = 110$: $A = a_1 = a_2 = 1$, $B = b_0 = b_1 = b_2 = b_3 = 1$, $C = c_1 = c_2 = c_3 = 0$, $g_1 = g_2 = g_{31} = g_{32} = g_4 = 1$, $Y = 1$, $Z = 0$. Simulating each single stuck-at (opposite) fault, the detected ones are

$$A/0,\ a_1/0,\ B/0,\ b_0/0,\ b_3/0,\ C/1,\ c_2/1,\ c_3/1,\ g_{31}/0,\ g_4/0,\ Y/0,\ Z/1$$

so $\boxed{12 \text{ faults}}$ are detected (13 if the unlabeled G3 output stem s-a-0 is counted). The faults $a_2/0, b_1/0, b_2/0, c_1/1, g_1/0, g_2/0, g_{32}/0$ are not detected (G3 is still kept at 1 by the other input, or Z is blocked by $C = 0$).

Problem 3. Find the test input when g_{31} is stuck-at 1 fault.

Solution

- **Activate:** g_{31} s-a-1 $\Rightarrow$ need $G3 = B + C = 0$, i.e. $B = 0,\ C = 0$.
- **Propagate:** through G5 needs $A = 1$ and $g_4 = B \oplus C = 1$, but $B = C = 0$ gives $g_4 = 0$. Contradiction.

Hence $\boxed{\text{no test input exists}}$: g_{31} s-a-1 is undetectable (redundant), since $Y = A(B+C)(B \oplus C) = A(B \oplus C)$ already does not depend on g_{31}.

Problem 4. What happens when applying the test input on 1 to the multiple faults of b_0 (s-a-0) and g_{31} (s-a-1)?

Solution

With $ABC = 110$ and both faults: $b_0 = 0 \Rightarrow g_1 = 0,\ g_2 = C = 0 \Rightarrow G3 = 0$, so $g_{32} = 0$, but g_{31} is stuck at 1. Then

$$Y = a_1 \cdot g_{31} \cdot g_4 = 1 \cdot 1 \cdot 1 = 1, \qquad Z = 0 \cdot 0 = 0,$$

which equals the good output. The undetectable g_{31} s-a-1 **masks** b_0 s-a-0, so the test 110 no longer detects b_0 s-a-0. In fact, with both faults $Y = A(B \oplus C)$ and $Z = C$, identical to the fault-free circuit, so this multiple fault cannot be detected by any input.